Problem 1a

Segment 1

$$\int_0^2 \sqrt{4-x^2}\,\mathrm{d}x$$

Problem 1b

Segment 1

$$\int_0^{\sqrt{2}} \sqrt{4-x^2}\,\mathrm{d}x$$

Problem 2a

Segment 1

$$\int \frac{\tan(2x)}{\tan(x)}\,\mathrm{d}x$$

Problem 2b

Segment 1

$$\int \frac{\tan(x)}{\tan(2x)}\,\mathrm{d}x$$

Problem 3a

Segment 1

$$\int_0^\infty \frac{1}{1+e^{\sqrt{x}}}\,\mathrm{d}x$$

Problem 3b

Segment 1

$$\int_0^\infty \frac{1}{1+(\sqrt{x})^e}\,\mathrm{d}x$$

Problem 4a

Segment 1

$$\int_0^\infty \sqrt{x^2\sqrt{2^{-x}\sqrt{x^4\sqrt{4^{-x}\sqrt{x^8\sqrt{8^{-x\sqrt{\cdots}}}}}}}}\,\mathrm{d}x$$

Problem 4b

Segment 1

$$\int_0^{\frac{1}{2}} \frac{x}{\sqrt{x+\frac{x}{\sqrt{x+\ldots}}}}+\sqrt{x^3+x^2\sqrt{x^3+x^2\sqrt{x^3+\ldots}}}\,\mathrm{d}x$$

Problem 5a

Segment 1

$$\int_{e^2}^{e^e} \frac{\ln\left(\frac{x^{(\ln(\ln(x)))^2}}{\ln(x)}\right)}{(\ln(\ln(x)))^3 \ln(x)} \, \mathrm{d}x$$

Problem 5b

Segment 1

$$\int_0^1 \ln\left(x^{\ln(1-\ln(x))+1-\ln(x)}\right) \, \mathrm{d}x$$

Problem 6a

Segment 1

$$\int_0^{\frac{\pi}{2}} \frac{\tan(\sin^2(x))}{1-\tan(\sin^2(x))\tan(\cos^2(x))} \, \mathrm{d}x$$

Problem 6b

Segment 1

$$\int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos(x)+\sin(x)}{\sqrt{2}\sin(x)}\right) + \ln(\tan(x)) \, \mathrm{d}x$$

Problem 7a

Segment 1

$$\int_0^1 \frac{1}{(1+x^2)(1+\frac{\pi}{2}+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x$$

Problem 7b

Segment 1

$$\int_0^1 \frac{1}{(1+x^2)(1+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x$$